

3D Coordinates Extension — Build the Spider (Intermediate)

Topic: 3D Spider (x, y, z) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Lesson 6) ·

Time: about 40 minutes

Build a **red spider** in 3D — a fat body, eight legs spread out, two eyes on the front. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

● Stand on your **home spot** (red block) every run. Move your feet, move your spider.

👁 Red parts = **red concrete**. Eyes = **black wool**.

1 · Predict 🧠

Read each set. Write a sentence for each.

```
place red block at (1, 1, 2)
place red block at (1, 1, 3)
place red block at (1, 1, 4)
```

Only z changes. What happens to this leg?


```
place red block at (1, 1, 2)
place red block at (5, 1, 2)
```

Only x changes. Where do these two legs sit?


```
place red block at (3, 2, 3)
place red block at (3, 3, 3)
```

Only y changes. What happens here?

2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

Bug A — This eye should sit **up high on the front**, at (2, 3, 3). It has only two numbers, so it drops flat on the floor.

```
place black block at (2, 3)
```

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences)

Bug B — This right-side leg should be **across on the right**: (5, 1, 3). The last two numbers are swapped.

```
place red block at (5, 3, 1)
```

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences)

3 · Show Your Work 📷 🎥

Look at the 🕷️ spider in the world. Walk around it. Then build **one full spider** on your home spot — body, four left legs, four right legs, two eyes.

Red body: fill from (2, 2, 3) to (4, 3, 4).

Left legs (share x = 1): red block at (1, 1, 2), (1, 1, 3), (1, 1, 4), (1, 1, 5).

Right legs (share x = 5): red block at (5, 1, 2), (5, 1, 3), (5, 1, 4), (5, 1, 5).

Two black eyes on the front: (2, 3, 3) and (4, 3, 3).

💡 Tip: legs reach **across** (x) to the sides and spread **deeper** (z) front to back. They stay low (small y).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera at the spider. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your **one** video to teacher on KakaoTalk.